

Senior High School



Lesson Exemplar in General Science

Quarter 2

LE

1

Lesson Exemplar for General Science
Quarter 2, Unit 1

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LESSON EXEMPLAR

Learning Area	General Science	Grade Level	11
Semester	1st	Quarter	2

I. OBJECTIVES (*Identifying the Goals*)

Content Standard	Learners learn that: Past research in the field of chemistry provides the foundations for the development of helpful products and processes;
Performance Standard	By the end of the quarter, learners apply research skills to find out how scientists have made an impact on our lives. They recognize the significant role of different agencies in the Philippines, such as DOST and DTI, in scientific innovation and development. They show critical thinking in explaining the different substances present in household and personal care products including their benefits and potential risk of using and disposing of them. They demonstrate appropriate and safe handling and use of chemicals. They explain that chemical reactions take place in human activities and in the environment, and that there are several chemical solutions that pose a threat to the community.
Learning Competencies	The learners... use information from secondary resources to report how Louis Pasteur combined research from the fields of chemistry and biology that has helped scientists to better understand the causes of some diseases and prevent infection;

II. REFERENCES and MATERIALS
(*Selecting Resources*)

References:

- Debré, P. (1998). *Louis Pasteur*. Johns Hopkins University Press.
- Geison, G. L. (1995). *The private science of Louis Pasteur*. Princeton University Press.
- Duclaux, É. (1920). *Pasteur: The history of a mind*. W.B. Saunders Company.
- Bulloch, W. (1938). *The history of bacteriology*. Oxford University Press.
- Brock, T. D. (1999). *Robert Koch: A life in medicine and bacteriology*. ASM Press.



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Department of Education
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and Material)	https://www.pasteur.fr/en/file/44615/download Materials: Journals, Books, illustrations, printed materials, laptops	
<i>(These shall be accomplished per topic)</i>		
III. CONTENT <i>(Sequencing Content)</i>	Chemists and how they have made an impact on our lives	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	At the end of the lesson, the learners will be able to: a. Create a timeline that highlights the major contributions of Louis Pasteur in the fields of chemistry and biology. b. Enumerate the contributions of each experiment and explain how they enhanced scientists’ understanding of the causes of disease and the prevention of infection. c. Identify modern products that were developed based on Pasteur’s scientific experiments. d. Explain how these products are applied in society today. e. Discuss the impact of Pasteur’s discoveries on daily life, particularly in the areas of medicine, food preservation, and commonly used materials. f. Recognize the role of chemists in improving the quality of human life.	
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i> (Each part shall have 2-3 varied activities)		ANNOTATION <i>*Instruction to teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge	VIRTUAL TOUR TO PASTEUR’S MUSEUM Students will be instructed to visit Pasteur’s Virtual Museum at https://artsandculture.google.com/story/XAXBwj6OxM9xKA	For the virtual museum tour activity, the teacher may request access to the computer laboratory, allowing students to explore Pasteur’s Museum more effectively.



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Prompt Questions

Pre-Tour Questions

1. What do you already know about Louis Pasteur and his contributions to science?
2. How do you think Pasteur's discoveries influence modern medicine and food safety?
3. What aspects of microbiology, vaccination, or pasteurization are you most curious about?

During the Tour Questions

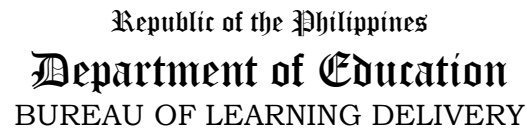
1. What scientific instruments or artifacts stand out to you? How do they compare to modern laboratory tools?
2. How did Pasteur's work contributed to public health regulations?
3. What challenges did Pasteur face in proving his theories, and how did he overcome them?
4. How does the museum present Pasteur's legacy in consumer education and regulatory knowledge?
5. What connections can you make between Pasteur's discoveries and FDA, DTI, or DENR regulations?

Post-Tour Reflection Questions

1. What was the most surprising fact you learned about Pasteur's work?
2. How can Pasteur's discoveries be applied to current scientific and regulatory challenges?

Prior to the virtual tour, the teacher may provide students with prompt questions to enhance their engagement and understanding.

If computer laboratories are unavailable, the teacher can still facilitate the museum experience by using a laptop connected to a larger television screen, ensuring all students have a better view of the exhibits.



WORD MATCH CHALLENGE

Description	Corresponding Word/s
1. Have you ever noticed that your left and right hands are similar yet not identical? This same characteristic is also found in molecules.	VACCINES GERM THEORY MOLECULAR ASYMMETRY FERMENTATION PASTEURIZATION INFECTIOUS DISEASE
2. What do yogurt, bread, and wine have in common? What causes milk to turn sour over time?	
3. It is what makes the milk you buy at the store safe to drink and helps it last longer in your fridge.	
4. "Why do we wash our hands or cover our mouths when we sneeze?"	
5. Colds or flu caught from someone else	
6. Shots given to you when you were a child to protect you from diseases.	

Word Match Challenge

Answers are:

MOLECULAR ASYMMETRY FERMENTATION
PASTEURIZATION GERM
THEORY INFECTIOUS
DISEASE VACCINES

Answer: Louis Pasteur

After answering the question, students will already have a clearer understanding of the lesson's content. This time, instruct the students to create a KWH chart and encourage them to respond to the questions in each column.

KWH Chart:

Responses that provide a broader perspective on Louis Pasteur are welcome, as this section is intended to evaluate the students' knowledge about Pasteur and his contributions.



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KWH-Chart. Students are expected to provide their answers to the questions listed under each column.

K (What do I know)	W (What do I want to know)	H (How do I find out)
K: What insights do you have regarding Louis Pasteur's contributions to society?	W: What aspects of his inventions and discoveries would you like to explore further?	H: How can you gain a deeper understanding of the contributions that chemists have made to society?

2. Establishing the Purpose of the Lesson

Activity: " **Pasteur's Breakthroughs: A Timeline Challenge** " —an interactive timeline activity where students will be grouped to become researchers uncovering Pasteur's groundbreaking achievements and create a timeline about his significant contributions in the field of chemistry.

Students may refer to the following sources.

Title	Type	URL	Short Description
<u>Louis Pasteur - Inventions, Achievements & Facts</u>	Article	bit.ly/4k8Le6E	This article explores the life and contributions of Louis Pasteur.
Louis Pasteur: Pioneer of Modern Medicine GCSE History	Video documentation	bit.ly/42UW5vj	The video showcases the groundbreaking contributions of Louis Pasteur to the field of medicine.

Pasteur's Breakthroughs: A Timeline Challenge

The following provides an overview of the significant milestones in the achievements of Louis Pasteur.

Date	Event
1854	Louis' becomes a professor of science Louis began his work as a professor, and one of his duties included finding scientific solutions to common problems in the area.
1855	Louis finds bacteria in the drink Louis was assigned the job of figuring out why common drinks like beer and wine became sour. He discovered the bacteria growing in the drinks and figured out that if you boiled the drinks and then cooled them, it killed the bacteria.
1864	He studies where the bacteria came from Pasteur went on to study where the bacteria growing in food and drinks were coming from, and he finally figured out it came from the air or environment.



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Department of Education
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	Biography: Louis Pasteur: A controversial figure in a debate on scientific ethics Author links open overlay panel B.Lee Ligon, PhD	Article	bit.ly/3GVpimgv	This article provides an overview of Pasteur's life and the debate regarding his scientific discoveries and honesty that has ensued for more than 100 years.	1864	The Germ Theory Many people in the 1800s still believed that germs were not harmful to people. Louis proved that germs were the cause of illness.
	The Remarkable Life and Legacy of Louis Pasteur	Video documentation	bit.ly/42KCLkg	The video chronicles the life and legacy of Louis Pasteur.	1865	Louis saves the silkworms After his success with the beverage industry, Louis was asked to work on the dying crops of local silk fields. He figured out the silkworms were ill and introduced a way to fix the problem.
	Students have the option to select their preferred method for presenting their timelines. They may choose from the following: a. Interactive Poster or Wall Display • Create a large, visually appealing timeline on a poster or bulletin board with images, dates, and short descriptions. • Use color-coding to differentiate periods or themes (e.g., scientific breakthroughs, personal milestones). b. Digital Timeline (Presentation or Infographic) • Use tools like PowerPoint, Canva, or Timeline JS to create an interactive digital timeline. • Add animations or clickable sections for a dynamic experience. c. Storyboard or Comic Strip • Represent key moments in Pasteur's life through illustrations, like a comic strip or storybook format.				1879	Louis discovers vaccinations Louis accidentally exposed chickens to a small form of cholera, and when the chickens didn't become sick, he wanted to know why. He figured out the small exposure had worked as a vaccine against the actual sickness, and the vaccine was born.
					1885	A vaccine against rabies After finding vaccines for smallpox, TB, and cholera, Louis worked toward a vaccine for rabies. He found it in 1885



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Department of Education
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	Students can narrate the discoveries using speech bubbles for each event.	"Pasteur's Breakthroughs: A Timeline Challenge". Students may work in groups for this activity; however, those who prefer to work independently may be permitted to do so. In this activity, the teacher can collaborate with the students to create rubrics, considering the following criteria. - Historical Accuracy - Creativity and Presentation - Organization & Clarity Additionally, the teacher may include other criteria based on the agreements made with the students.
B. Instituting New Knowledge	1. Presenting Examples Select a group or an individual to present their created timeline. From the presentations of students, give emphasis on the following: Pasteurization in Food Safety – His pasteurization process is still used today to kill harmful bacteria in milk, juice, and other beverages, ensuring they are safe for consumption. Vaccines for Disease Prevention – Pasteur's work on vaccines led to the development of life-saving immunizations, including those for rabies and anthrax, paving the way for modern vaccination programs. Improved Hygiene and Sanitation – His germ theory of disease revolutionized medical practices, leading to better sterilization techniques in hospitals and reducing infections.	At this stage, students should grasp that Pasteur's experiments established the groundwork for the discovery of numerous products that society benefits from today. After the presentation, the teacher may give emphasis that Pasteur's experiments have had profound impact on modern science and daily life. The teacher may also cite supplementary sources to offer a more comprehensive understanding of Pasteur's experiments.



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Department of Education
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4. **Advancements in Microbiology** – His discoveries laid the foundation for microbiology, helping scientists understand how microorganisms affect health and industry.
5. **Protection of the Wine and Silk Industries** – Pasteur's research saved the French wine industry from spoilage and helped prevent diseases in silkworms, preserving economic stability.

2. Discussing New Concept:

Activity: "Pasteur's Scientific Solutions"

Divide the class into teams. Each team represents a group of scientists facing a mysterious disease outbreak. Their goal is to use Pasteur's discoveries to resolve the following problems.

1. Germ Theory Challenge

A hospital is struggling with patients getting infections after surgery. Some doctors believe diseases come from "bad air." What experiment from Pasteur could help prove the real cause?

2. Pasteurization Challenge

Farmers in a village complain that their milk spoils too quickly, making people sick. What scientific process can they use to keep milk safe for longer?

3. Rabies Vaccine Challenge

A child was bitten by a rabid dog, the Barangay health workers who are supposed to provide first aid do not know how to save him. What method, based on Pasteur's work, could help protect people from the deadly rabies?

Activity: "Pasteur's Scientific Solutions"

The activity will encourage students to utilize their knowledge, think critically, and interact with Pasteur's discoveries in an enjoyable manner! The answers provided below can serve as a foundation for the teacher in evaluating their responses. May also add other details.

Expected Answers:

1. Germ Theory Challenge

Pasteur's germ theory experiment demonstrated that microorganisms cause disease. His findings led to the importance of sterilization in hospitals to prevent infections.

2. Pasteurization Challenge

Pasteurization—Pasteur discovered that heating liquids to a specific temperature could kill harmful bacteria, preventing foodborne illnesses.

3. Rabies Vaccine Challenge



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Department of Education
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4. Spontaneous Generation Challenge Some scientists argue that bacteria appear randomly in spoiled food. What experiment could prove that microorganisms come from other living things rather than just appearing out of nowhere? 3. Developing Mastery Pasteur's Scientific Impact Symposium Students will be grouped in a roundtable discussion format, as if they are scientists presenting/discussing their research on the following topics. Each discussion will be guided by the question anchored in each topic. 1. Historical Significance - How did Pasteur's experiments change scientists' understanding of disease prevention? - Which of his contributions do you think had the most lasting impact on modern medicine? Defend your answer. 2. Ethical & Societal Impact - How did Pasteur's discoveries affect public health policies? Give examples of laws or practices influenced by his work. - If Pasteur had not conducted his experiments, how might modern healthcare look different today? 3. Connection to Modern Science - What scientific advancements today build on Pasteur's work (e.g., vaccines, antibiotics, food safety)?	Pasteur's rabies vaccine—His research on weakened viruses led to the development of vaccines that protect against rabies and other diseases. 4. Spontaneous Generation Challenge Pasteur's swan-neck flask experiment demonstrated that microorganisms come from the environment, disproving spontaneous generation. Pasteur's Scientific Impact Symposium The activity promotes critical thinking, thorough research, and well-structured oral presentations instead of mere memorization. - Focuses on real-world applications of historical discoveries. - Prepares students for deeper discussions and scientific analysis. The teacher may agree with students on the rubrics to be used in the activity. May consider the following criteria: - Understanding of Pasteur's Contributions - Critical Thinking & Application - Clarity & Organization - Engagement in Discussion / Argumentation - Use of Research & Evidence Students can select from the available activity options. They may collaborate as a group; however, if someone prefers to work independently, they are permitted to do so.
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Department of Education
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	- Compare and contrast the challenges Pasteur faced in his time with challenges scientists face today when making medical discoveries. 4. Application Scenarios - Imagine there is a new infectious disease outbreak. Based on Pasteur's principles, what would be the first steps in controlling the spread? - If Pasteur were alive today, what emerging scientific problems do you think he would work on?	
C. Demonstrating Knowledge and Skills	Finding Practical Application Activity: "Pasteur's Legacy – Then & Now" Objective: Students will explore how Pasteur's discoveries have influenced modern products and conducting research and presenting real-world applications. Students may choose from the following options Product Investigation Students will be assigned the task of tracing the origins of various modern products, such as mRNA vaccines, high-pressure pasteurization, probiotics, and antimicrobial coatings, back to the foundational experiments of Pasteur. This includes exploring concepts like germ theory, fermentation, and vaccine development.	Activity: "Pasteur's Legacy – Then & Now" Allow students to choose among the activities, which may be done by group or individually, depending upon the student's preference. The teacher may require students to conduct online research or may provide required readings and references. Rubrics may be done based on agreements made with students.



Republic of the Philippines
Department of Education
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	2. Innovation Showcase Create mini-presentations (posters, slides, or short videos) explaining how Pasteur's work led to the development of their assigned product. Include scientific connections, benefits, and real-world impact. 3. Debate & Discussion. The debate will focus on the question: "Which of Pasteur's discoveries has had the greatest impact on modern science?" Students defend their assigned product using historical and scientific reasoning. 4. Creative Extension – "Future Innovations" Ask students to predict future advancements based on Pasteur's principles. What new medical or food safety technologies might emerge in the next decade? 2. Making Generalization Personal Reflection Students should think about these questions and give thoughtful answers. 1. Why do scientists need to combine multiple fields of study (like chemistry, biology and technology) to advance medicine?	In making generalizations, the teacher may assign one question to each group or may invite volunteers to respond to the posted questions. The teacher can emphasize the positive aspects of the students' answers
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2. How does Pasteur's legacy inspire future innovations in healthcare?
3. Imagine a world **without Pasteur's discoveries**. How would hospitals, food safety, and disease prevention look different today?
4. How does Pasteur's work help scientists combat **new diseases**, such as COVID-19?

3. Evaluating Learning

Summative Exam: The Legacy of Louis Pasteur

I. Multiple Choice (Choose the Best Answer) (2 points each)

1. What theory did Pasteur's **swan-neck flask experiment** disprove?
 - a. Germ theory
 - b. Miasma theory
 - c. Spontaneous generation
 - d. Cell theory
2. Pasteur's discoveries contributed significantly to which of the following?
 - a. Development of antiseptics
 - b. The invention of the telescope
 - c. Discovery of DNA structure
 - d. Creation of the periodic table

and provide additional details to enhance the responses when necessary.

Summative Exam: The Legacy of Louis Pasteur

Answer Key & Explanations. Teacher may briefly discuss concepts when checking the summative assessment. May add details to the given explanations/discussions below.

I. Multiple Choice (2 pts each)

1. (c) Spontaneous generation
 - o Pasteur's swan-neck flask experiment disproved the long-held belief that microorganisms appeared randomly. Instead, he showed that they came from external contamination.
2. (a) Development of antiseptics
 - o Pasteur's germ theory led scientists like Joseph Lister to develop antiseptic



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3. Which industry benefited **directly** from Pasteur's pasteurization process?
 - a. Textile industry
 - b. Dairy and food industry
 - c. Automobile industry
 - d. Electronics industry
4. Which of the following **best explains** why Pasteur's germ theory changed medical practices?
 - a. It proved disease was caused by tiny organisms, leading to better hygiene and sterilization.
 - b. It showed how plants grow faster using fertilizers.
 - c. It introduced electricity as a treatment for infections.
 - d. It recommended eating raw foods to improve immunity.
5. Pasteur's vaccine development laid the foundation for which **modern medical advancement**?
 - a. Ultrasound technology
 - b. Antibiotic discovery
 - c. mRNA vaccines
 - d. Blood transfusions

practices, which revolutionized surgery and hospital sanitation.

3. (b) Dairy and food industry
 - Pasteurization is widely used to keep milk, juices, and other food products safe by eliminating harmful bacteria.
4. (a) It proved disease was caused by tiny organisms, leading to better hygiene and sterilization.
 - Before Pasteur, people believed illnesses were caused by "bad air" (miasma theory). His discoveries led to sterilization, vaccination, and improved public health practices.
5. (c) mRNA vaccines
 - Pasteur's vaccine research paved the way for modern mRNA vaccines used today for diseases like COVID-19.

II. Matching Type (2 pts each)

Experiment	Contribution
Pasteurization	Prevents bacterial growth in food and beverages
Germ Theory	Led to sterilization techniques in hospitals



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II. Matching Type (2 points each)

Match Pasteur's Experiments to their corresponding Contributions in the second column.

Experiment	Contribution
Pasteurization	Proved microorganisms come from the environment
Germ Theory	Led to sterilization techniques in hospitals
Rabies Vaccine	First vaccine developed for a viral disease
Spontaneous Generation Disproof	Prevents bacterial growth in food and beverages

4. Additional Activities

Short Answer (5 pts each)

1. Explain how Pasteur's discovery of germ theory helped prevent infections in hospitals.
2. Describe a modern application of Pasteur's fermentation research in food or medicine.
3. What would happen to public health if Pasteur's discoveries had never been made?
4. How did Pasteur's rabies vaccine experiment change the way vaccines are developed today?
5. What are the differences between spontaneous generation and Pasteur's germ theory?

Rabies Vaccine	First vaccine developed for a viral disease
Spontaneous Generation Disproof	Proved microorganisms come from the environment

Discussion:

These experiments all contributed to modern medicine, food safety, and hygiene. Pasteur's work laid the foundation for today's healthcare advancements and bacterial research.

Short Answer (5 pts each). The teacher may use the following brief responses as basis in giving points to the students' responses.

1. Hospital infection prevention

Pasteur's germ theory proved that bacteria and viruses cause disease, leading to handwashing, sterilization, and improved hospital hygiene, drastically reducing deaths from infections.

2. Fermentation applications

Pasteur's research on fermentation influenced probiotic development, antibiotic production, and food safety techniques, helping industries improve digestive health and preservation methods.

3. Life without Pasteur's discoveries



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		Without Pasteur's work, hospitals would lack sterilization protocols, vaccines wouldn't exist, and foodborne illnesses would be much more common, leading to higher mortality rates. 4. Rabies vaccine advancements Pasteur's rabies vaccine was the first successfully developed vaccine for a viral disease, inspiring modern vaccine technologies like flu shots, HPV vaccines, and mRNA advancements. 5. Spontaneous generation vs. germ theory Spontaneous generation falsely suggested that life appeared out of nowhere, while germ theory proved that microorganisms come from contamination, leading to infection control improvements.						
V. ASSESSMENT <i>(Assessing Learnings)</i>	Students will fill individually answer the questions in the LAQ Chart. Teacher may give score to the students' responses based on the suggested rubrics. <table border="1"> <thead> <tr> <th>L(What have I learned)</th><th>A (What Actions Will I take)</th><th>Q (What New questions Do I have?)</th></tr> </thead> <tbody> <tr> <td> - How do Pasteur's contributions inspire future scientists and medical researchers? - If Pasteur had not conducted his experiments, how would our world be different today? </td><td> - How can you apply Pasteur's discoveries in your daily life to promote hygiene and disease prevention? </td><td> - If you were a scientist building on Pasteur's work, what new discoveries or improvements would you focus on? </td></tr> </tbody> </table>		L(What have I learned)	A (What Actions Will I take)	Q (What New questions Do I have?)	- How do Pasteur's contributions inspire future scientists and medical researchers? - If Pasteur had not conducted his experiments, how would our world be different today?	How can you apply Pasteur's discoveries in your daily life to promote hygiene and disease prevention?	If you were a scientist building on Pasteur's work, what new discoveries or improvements would you focus on?
L(What have I learned)	A (What Actions Will I take)	Q (What New questions Do I have?)						
- How do Pasteur's contributions inspire future scientists and medical researchers? - If Pasteur had not conducted his experiments, how would our world be different today?	How can you apply Pasteur's discoveries in your daily life to promote hygiene and disease prevention?	If you were a scientist building on Pasteur's work, what new discoveries or improvements would you focus on?						



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- How does learning about Pasteur's work help you appreciate the role of science in improving lives?

Suggested rubrics in Giving scores

Criteria	Excellent (4 pts)	Proficient (3 pts)	Developing (2 pts)	Needs Improvement (1 pt)
Learning (L) - Understanding Pasteur's Contributions	Provides a detailed, insightful analysis of Pasteur's discoveries and their impact on science, medicine, and public health.	Explains Pasteur's contributions well with clear connections to scientific advancements.	Provides basic information on Pasteur's work but lacks depth or historical connections.	Minimal understanding of Pasteur's discoveries; lacks relevance or clarity.
Action (A) - Applying Pasteur's Discoveries in Daily Life	Suggests meaningful and practical applications for hygiene, disease prevention, or modern scientific advancements.	Provides good examples of hygiene and safety practices inspired by Pasteur but lacks innovation.	Identifies some ways Pasteur's discoveries apply to daily life but lacks clarity.	Limited or vague ideas on how Pasteur's work relates to personal or public health practices.
Question (Q) - Generating Further Inquiry	Asks thought-provoking, research-worthy questions.	Proposes good questions that explore new scientific.	Provides basic questions but lacks depth or connection to future.	Does not ask meaningful or relevant questions.



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**VI.
REFLECTI
ON**
(Feedback
and
Continuou
s
Improveme
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